

Thesis Index

Introduction

Introduction to Jailbreaking and LLM Safety

Defining the phenomenon and the importance of safety in Large Language Models

Research objectives and methodology

1. Jailbreak attacks on public repositories

1.1 Common standards: Harmbench and WildJailbreak

Analysis of reference safety benchmarks

1.2 Multilingual Safety Evaluation

Disparities in defenses across different languages: the "Western Shield" concept

1.3 Arabic and Russian datasets focus

Specific analysis of vulnerabilities in low-resource languages or non-Latin scripts

1.4 Key Takeaways from Public Repositories of prompts

Synthesis of learned lessons and limitations of public datasets

2. Jailbreak attacks on AI-generated prompts

2.1 Topic Selection and Customization of Prompts

Strategies for selecting risk domains (Chemical, Cybercrime, Harassment)

2.2 Automated Adversarial Prompt Generation

Synthetic generation methodologies: from prompt engineering to automated LLM-based Red Teaming

2.3 Comparative Analysis: Human-crafted vs. AI-generated Jailbreaks

Evaluating efficacy and sophistication between repository-sourced and synthetically generated prompts

3. Development of a predictive model for Jailbreak risk assessment

3.1 Feature Engineering: Textual and Semantic Analysis Metrics

The role of Toxicity Score, linguistic complexity (Perplexity), and Token Length in predicting attack success

3.2 Machine Learning Model Architecture and Training

Selection of the model and training on success/failure datasets

3.3 System Implementation and User Interface Design

Development of the Backend-Frontend architecture: integrating the ML model for real-time assessment of compromise probability

Conclusions and Future Developments

Summary of results obtained

Perspectives on the resilience of next-generation models